

Course: B. Tech.

Semester : VII

Branch: Mechanical Engineering/ Mechanical Engineering (Sandwich)/Production Engineering.

Subject Code & Name: BTMPE703C Non -Conventional Machining

Max Marks: 60

Date: 05/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) How the unconventional machining processes are classified.	CO1	6
B) Justify the need of non-conventional machining in today's Industries.	CO1	6
C) Discuss the limitations of conventional machining processes.	CO2	6
Q.2 Solve Any Two of the following.		12
A) What are the specific advantages of using chemical machining over electrochemical machining.	CO1	6
B) Differentiate between cut and peel resists and photographic resists.	CO2	6
C) Explain electro chemical deburring and honing process.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Discuss merits of EDM with respect to MRR and surface finish.	CO4	6
B) Explain the nontransferred and transferred mode of plasma arc.	CO2	6
C) Explain electron beam machining with neat sketch.	CO2	6
Q.4 Solve Any Two of the following.		12
A) Give the practical applications of water jet machining	CO2	6
B) Discuss why the AJM technique when applied to ductile materials lead to a low rate of material removal.	CO4	6
C) Explain the tool feed mechanism used in ultrasonic machining process.	CO2	6
Q. 5 Solve Any Two of the following.		12
A) Compare the laser beam and electron beam machining processes.	CO5	6
B) Classify the Hybrid Machining process.	CO5	6
C) Describe the electro discharge chemical grinding with neat sketch.	CO6	6